

STAAD.Pro

Software Course | Duration: 8 Weeks

A structural analysis and design course covering modeling, loading, code-based analysis, and RC/steel member design for multi-storey buildings using STAAD.Pro — the industry-standard tool for structural consultants.

Who Should Enroll: Civil engineering students, structural design trainees, and site/design engineers moving into structural analysis roles.

Prerequisites

- Basic structural analysis / engineering mechanics
- Familiarity with IS 456 and IS 800 is helpful

Learning Outcomes

- Model 3D framed structures with correct geometry, sections, and supports
- Apply dead, live, wind, and seismic loads as per IS 875 and IS 1893
- Interpret analysis results — BM, SF, deflection, and reactions
- Design RC beams/columns/footings and steel members within the software
- Produce a complete design report for a real building frame

Week-by-Week Curriculum

Week	Focus	Topics Covered
Week 1	Modeling Basics	– Interface, units, and coordinate systems – Creating nodes, beams, and plate geometry – Assigning material and section properties
Week 2	Supports & Geometry Practice	– Support conditions: fixed, pinned, roller, spring – Generating multi-storey geometry using repeat/translate tools
Week 3–4	Loading & Combinations	– Dead, live, wind (IS 875) and seismic (IS 1893) load definition – Load combinations and load generation for lateral cases – Running analysis and diagnosing model/analysis errors
Week 5	Result Interpretation	– Reading BM, SF, deflection, and reaction outputs – Cross-checking software output against manual estimates
Week 6–7	Design Modules	– RC design of beams, columns, and footings (IS 456) – Steel member design (IS 800) – Response spectrum / dynamic analysis basics
Week 8	Capstone Project	– End-to-end analysis and design of a multi-storey building frame – Design report generation and BBS-ready output – Review session and Q&A on real project queries

Capstone Project

Full structural analysis and design of a multi-storey RC/steel building frame, with a generated design report.

Certification: On successful completion, learners receive the VCTA Certificate of Completion, along with a portfolio-ready project file that can be showcased to employers or clients.